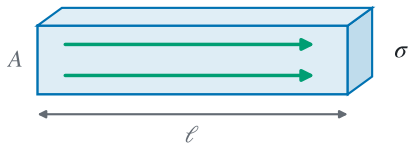
Electric circuit: a conductor confines current

$\mathbf{J} = \sigma \mathbf{E}$, total $I = JA$



$$V = \int \mathbf{E} \cdot d\boldsymbol{\ell} = I \frac{\ell}{\sigma A} = RI$$

$\nabla \cdot \mathbf{J} = 0 \Rightarrow \sum I_k = 0$ (node)

$\oint \mathbf{E} \cdot d\boldsymbol{\ell} \Rightarrow \sum R_k I_k = \sum V_s$ (loop)

air is an excellent insulator

$\sigma_{\mathrm{Cu}}/\sigma_{\mathrm{air}} \gtrsim 10^{21}$: leakage current negligible

$\mathbf{E}$	↔	$\mathbf{H}$
$\mathbf{J}$	↔	$\mathbf{B}$
σ	↔	μ
V	↔	$\mathcal{F}$
I	↔	Φ
R	↔	$\mathcal{R}$
G	↔	$\mathcal{P}$

same form,
not same
scale

Magnetic circuit: iron confines flux

$\mathbf{B} = \mu \mathbf{H}$, total $\Phi = BA$

$$\int \mathbf{H} \cdot d\boldsymbol{\ell} = \Phi \frac{\ell}{\mu A} = \mathcal{R}\,\Phi$$

$\nabla \cdot \mathbf{B} = 0 \Rightarrow \sum \Phi_k = 0$ (node)

$\oint \mathbf{H} \cdot d\boldsymbol{\ell} = I_f \Rightarrow \sum \mathcal{R}_k \Phi_k = \sum \mathcal{F}_j$ (loop)

flux has no insulator

$\mu_{\mathrm{Fe}}/\mu_0 \sim 10^3$–$10^5$: leakage often matters